MEASURABLE DENSITY BOUNDS HOLD FOR ALL COLOURINGS OF THE PLANE WITH ALGEBRAIC FORBIDDEN DISTANCES

SERGI GALÁN

Abstract. OpenAI recently proved that every colouring of the plane with five colours gives the same colour to two points at distance 1. The proof transfers an arbitrary proper colouring of the plane to a *weak measurable* one, and then excludes weak measurable 5-colourings. We observe that the transfer works, with the same proof, for every set of algebraic forbidden distances and for fractional colourings. Upper bounds for the density of measurable sets avoiding given distances therefore become lower bounds for arbitrary colourings; all these lower bounds rest on OpenAI's rigidity theorem, which OpenAI formalized in Lean; we formalized the transfer for proper colourings (not its fractional version) on top of their formalization. Consequences: the fractional chromatic number of the plane equals $1/m_1(\mathbb{R}^2)$, where $m_1(\mathbb{R}^2)$ is the largest density of a measurable planar set with no two points at distance 1, so it lies between 4.0485 and 4.36; and the least independence ratio of an n-point unit-distance graph tends to $m_1(\mathbb{R}^2)$, as conjectured by Dúcz and Varga. If the distances 1 and 2 are both forbidden, at least 7 colours are needed, improving the bound 6 of Exoo and Ismailescu; the same holds for $\{1, \sqrt{3}\}$, and at least 8, 10, 11 and 13 colours are needed for $\{1, 11/6\}$, $\{1, 40\}$, $\{1, 100\}$ and $\{1, 1000\}$. Uniformly in d, $\chi(\mathbb{R}^2, \{1, d\}) \geq 7$ for every real algebraic $d \geq 5.82$ and for $1.72 \leq d \leq 2.04$, and the bound rises step by step to 14 for every real algebraic $d \geq 25\,700$, 15 for $d \geq 5 \cdot 10^5$, 16 for $d \geq 7 \cdot 10^7$ and 17 for $d \geq 8 \cdot 10^8$; each step is proved by a single certificate for a whole half-line of d, using unit equilateral triangles, Moser spindles and, from 15 colours on, the 23-point inclusion–exclusion program of Ambrus, Csiszárik, Matolcsi, Varga and Zsámboki at the scale of 1. For 16 and 17 a second copy of that program bounds, at the scale of d, the set of points near which the set meets a copy of their configuration; this makes the bounds at the two scales multiply, with explicit thresholds, as the optimal densities do in the limit by Bukh's theorem. The distances 1, 100 and 10^4 force 27 colours, 1, 10, 100 and 1000 force 26, and 1, 30, 900 and 27000 force 48. For suitable sets of k integer distances more than 4.0485^k colours are needed, in the plane and in every higher dimension, so the chromatic number with k forbidden distances grows exponentially in k, whereas Erdős expected polynomial growth (this answers Erdős Problem #706). Davies has just proved the analogous statements in dimension at least 4 by a different method. Using OpenAI's theorem itself at a small scale and density bounds at a large one, we show that the distances 1 and N force at least 25 colours for every large algebraic N, although 19.01 colours suffice fractionally, and, with de Grey's graph, at least 18 colours for every algebraic $N \geq 10^{25}$; iterating, k distances can force at least $6 \cdot 4.0486^{k-1}$ colours. The density bounds we use are linear-programming bounds of Oliveira Filho and Vallentin, which we certify in interval arithmetic, and theorems of Bukh and of Ambrus, Csiszárik, Matolcsi, Varga and Zsámboki.

1. Results

For a set D of positive real numbers, $\chi(\mathbb{R}^2, D)$ is the least number of colours needed to colour the plane so that no two points whose distance lies in D get the same colour. The Hadwiger–Nelson problem is the case $D = \{1\}$; OpenAI [23] proved $\chi(\mathbb{R}^2) \geq 6$, and $\chi(\mathbb{R}^2) \leq 7$ is Isbell's hexagonal colouring [28]. For two distances $\{1, d\}$ the best known lower bound was 6: Owings, Tetiva and Huddleston [24] showed $\chi(\mathbb{R}^2, \{1, (1+\sqrt{5})/2\}) \geq 6$ (as reported in [4, 10]), Exoo and Ismailescu [13] proved $\chi(\mathbb{R}^2, \{1, 2\}) \geq 6$, Parts [25] gave a small 6-chromatic two-distance

Date: Draft, version 8, 11 October 2026.

2020 *Mathematics Subject Classification.* Primary 05C15, 52C10; Secondary 05C72, 42B10, 90C05.

Key words and phrases. Hadwiger–Nelson problem, chromatic number of the plane, forbidden distances, fractional chromatic number, distance-avoiding sets, linear programming bounds.

These results were obtained with the assistance of an AI system (Claude, by Anthropic), which found the observation, wrote the programs, ran the checks and helped to draft this note. Seven separate AI agents refereed the note with their own programs. The hard input is OpenAI's transfer theorem [23]. No mathematician has checked the note yet. The author is not a professional mathematician and submits it for independent verification.

graph, and De Neve, Vanden Kerchove, Colle, Tavernier and Pickavet [10] gave 6-chromatic graphs for $d = \sqrt{3}$ and $d = (\sqrt{6} + \sqrt{2})/2$, two cases claimed earlier in the Polymath16 project [26]; Chybowska-Sokół, Junosza-Szaniawski and Węsek [4] record as open whether some pair $\{1, d\}$ forces 7 colours.

OpenAI's proof has two steps. Their Theorem 1.3 (the transfer) says that the plane has a proper k-colouring if and only if it has a *weak measurable* k-colouring: a measurable colouring in which the pairs of points at distance 1 with the same colour have measure zero. Their Theorem 1.4 excludes weak measurable 5-colourings. The transfer rests on a rigidity theorem for invariant measures on the character group of the algebraic plane [23, Thm. 2.3]. Its proof uses neither that the forbidden distance is 1 nor that a colouring uses finitely many colours rather than being fractional. We record the two generalisations (Theorems 12 and 13) and their consequences.

Write $m_D(\mathbb{R}^2)$ for the supremum of the upper densities of measurable sets $A \subseteq \mathbb{R}^2$ with no two points at a distance in D, and $m_1(\mathbb{R}^2) = m_{\{1\}}(\mathbb{R}^2)$. Croft's construction gives $m_1(\mathbb{R}^2) \geq 0.229364$ [6], and Ambrus, Csiszárik, Matolcsi, Varga and Zsámboki [1] proved $m_1(\mathbb{R}^2) \leq 0.2470$, confirming a conjecture of Erdős that $m_1(\mathbb{R}^2) < 1/4$. For the fractional chromatic number we use two versions, defined in Section 2: the finitary one $\chi_{\mathrm{f},0}(\mathbb{R}^2, D)$, the supremum over finite graphs drawn in the plane with edge lengths in D, and the fractional chromatic number $\chi_{\mathrm{f}}(\mathbb{R}^2, D)$ of the whole infinite graph.

Theorem 1. *Let D be a finite set of positive real algebraic numbers. Then*

$$\chi(\mathbb{R}^2, D) \;\geq\; \chi_{\mathrm{f},0}(\mathbb{R}^2, D) \;=\; \chi_{\mathrm{f}}(\mathbb{R}^2, D) \;=\; \frac{1}{m_D(\mathbb{R}^2)}.$$

The lower bound $\chi_{\mathrm{f},0}(\mathbb{R}^2, D) \geq 1/m_D(\mathbb{R}^2)$, and with it every lower bound below, rests on OpenAI's rigidity theorem [23, Thm. 2.3], which OpenAI formalized in Lean; the upper bound $\chi_{\mathrm{f}}(\mathbb{R}^2, D) \leq 1/m_D(\mathbb{R}^2)$ is elementary.

One distance.

Corollary 2. $\chi_{\mathrm{f},0}(\mathbb{R}^2) = \chi_{\mathrm{f}}(\mathbb{R}^2) = 1/m_1(\mathbb{R}^2)$, *and* $4.0485 < 1/0.2470 \leq \chi_{\mathrm{f}}(\mathbb{R}^2) \leq 1/0.229364 < 4.36$.

The constants in [1, §8] give $m_1(\mathbb{R}^2) < 0.2469975$, hence $\chi_{\mathrm{f}}(\mathbb{R}^2) > 4.0486$. Matolcsi, Ruzsa, Varga and Zsámboki [20] proved $\chi_{\mathrm{f}}(\mathbb{R}^2) \geq 4$ and conjectured that $\chi_{\mathrm{f},0}(\mathbb{R}^2) = 4$ and that $\chi_{\mathrm{f}}(G) < 4$ for every finite unit-distance graph G [20, Conj. 1]; the value 4 had also been conjectured in [1, Remark after Cor. 2]. Dúcz and Varga [12] disproved this conjecture: they found a unit-distance graph G_{29} whose geometric fractional chromatic number exceeds 4.0007, which gives a finite unit-distance graph with independence ratio below $1/4$ and $\chi_{\mathrm{f}}(\mathbb{R}^2) > 4.0007$. Corollary 2 gives a second disproof and determines $\chi_{\mathrm{f}}(\mathbb{R}^2)$ in terms of $m_1(\mathbb{R}^2)$. It also answers the question of [20] whether $\chi_{\mathrm{f},0}(\mathbb{R}^2) = \chi_{\mathrm{f}}(\mathbb{R}^2)$, and with [20, Thm. 2] it shows that the finitary independence ratio of the plane equals the measurable one, which [20] remark would fail if their conjecture held. Let $f(n)$ be the least independence number of the unit-distance graph of a set of n points in the plane, the function of Erdős's question [16] answered by Dúcz and Varga.

Corollary 3. $f(n) \geq m_1(\mathbb{R}^2)\, n$ *for every n, and* $f(n)/n \to m_1(\mathbb{R}^2)$ *as* $n \to \infty$. *In particular* $f(n) < 0.2470\, n$ *for all large n.*

The first statement is the classical averaging bound $m_1(\mathbb{R}^2) \leq \alpha_1(\mathbb{R}^2)$ [1, Remark after Cor. 2], [20, (14)], [12, Claim 1]; the new content is the limit, that is, $\alpha_1(\mathbb{R}^2) = m_1(\mathbb{R}^2)$. Dúcz and Varga conjectured that $f(n)/n = m_1(\mathbb{R}^2) + o(1)$, and in particular that $m_1(\mathbb{R}^2)$ is the density $0.2293647\ldots$ of Croft's construction [12, Conj. 1]. Corollary 3 proves the first statement; the second remains open.

Two distances.

Corollary 4.

$$\chi(\mathbb{R}^2,\{1,2\})\ge 7,\qquad \chi(\mathbb{R}^2,\{1,\sqrt{3}\})\ge 7,\qquad \chi(\mathbb{R}^2,\{1,(\sqrt{6}+\sqrt{2})/2\})\ge 7,$$
$$\chi(\mathbb{R}^2,\{1,11/6\})\ge 8,\qquad \chi(\mathbb{R}^2,\{1,40\})\ge 9,$$
$$\chi(\mathbb{R}^2,\{1,100\})\ge 10,\qquad \chi(\mathbb{R}^2,\{1,1000\})\ge 12.$$

Moreover $\chi(\mathbb{R}^2,\{1,d\})\ge 17$ *for every sufficiently large real algebraic number* d.

The finite bounds come from linear programs certified in Section 6 (Theorem 5 below raises those for $\{1,40\}$, $\{1,100\}$ and $\{1,1000\}$ to 10, 11 and 13), and the last statement from Bukh's theorem [2] (Theorem 5 gives it for every $d\ge 8\cdot 10^8$); products of hexagonal colourings give $\chi(\mathbb{R}^2,\{1,d\})\le 49$. For *measurable* colourings, bounds of this kind were known: Oliveira Filho and Vallentin [22, §4] report, from their linear program, the bound $0.141577<1/7$ for the density of a measurable set avoiding the distances 1 and $j_{1,2}/j_{1,1}\approx 1.8309$, a ratio of zeros of J_1, and, combining Bukh's theorem with their bound for $m_1(\mathbb{R}^2)$, they show $\inf_{d_1,d_2} m_{\{d_1,d_2\}}(\mathbb{R}^2)\le m_1(\mathbb{R}^2)^2\le 0.072046$; so pairs of distances whose density is close to $m_1(\mathbb{R}^2)^2$ need 14 colours for measurable colourings (17 with the bound of [1]). What is new here is that such bounds hold for arbitrary colourings; the bounds for $\{1,2\}$ and $\{1,\sqrt{3}\}$, which use clique inequalities, also seem to be new for measurable colourings.

The same linear program gives bounds for whole ranges of d at once.

Theorem 5. *Let* $d\ge 1$ *be a real algebraic number and* $7\le k\le 17$. *Then* $\chi(\mathbb{R}^2,\{1,d\})\ge k$ *whenever* $d\ge d_k$, *where*

$$d_7=5.82,\quad d_8=10.5,\quad d_9=18.9,\quad d_{10}=35.7,\quad d_{11}=73.6,\quad d_{12}=181,$$
$$d_{13}=655,\quad d_{14}=25\,700,\quad d_{15}=5\cdot 10^5,\quad d_{16}=7\cdot 10^7,\quad d_{17}=8\cdot 10^8.$$

Moreover $\chi(\mathbb{R}^2,\{1,d\})\ge 7$ *for* $1.72\le d\le 2.04$, *and* $\chi(\mathbb{R}^2,\{1,d\})\ge 8$ *for* $1.82\le d\le 1.89$. *The same bounds hold for* $\chi(\mathbb{R}^2,\{1,1/d\})=\chi(\mathbb{R}^2,\{1,d\})$, *and, for measurable colourings, for every real* d.

Theorem 5 contains the finite bounds of Corollary 4 and raises the ones for $\{1,40\}$, $\{1,100\}$ and $\{1,1000\}$ to 10, 11 and 13; see Figure 1. Scaling the plane by $1/d$ turns $\{1,d\}$ into $\{1/d,1\}$, which gives the equality with $1/d$. For measurable colourings neither the transfer nor algebraicity is needed: Proposition 18 and Lemma 19 do not use them, and one class of a measurable proper k-colouring avoiding $\{1,d\}$ is a measurable set avoiding $\{1,d\}$ of upper density at least $1/k$. For measurable colourings bounds of this size were known for one pair [22, §4] and, by Bukh's theorem, for all d beyond an unspecified threshold (see the discussion after Corollary 4); there the new content is the explicit thresholds and ranges. For each k a single function serves every $d\ge d_k$ (Lemma 19 and Table 3), and each of the two bounded ranges is covered by finitely many functions, each serving an interval of d (Section 7). Up to 14 colours they use the clique inequalities of unit equilateral triangles, the tails also the inequalities of Moser spindles (Proposition 18 with $\alpha=2$, as in [19]); from 15 colours on they use the inclusion–exclusion program of [1] at the scale of 1 (Proposition 20), and for 16 and 17 also at the scale of d, applied to the set of points near which the set meets a copy of their configuration (Proposition 22). Bukh's theorem gives 17 colours beyond an unspecified threshold, and Theorem 9 with de Grey's graph gives 18 for $d\ge 10^{25}$ (Remark 5). For 15, 16 and 17 colours Theorem 5 lowers that threshold to $5\cdot 10^5$, $7\cdot 10^7$ and $8\cdot 10^8$, and, unlike Theorem 9, it rests on density bounds: for these d it also gives $\chi_{\mathrm{f}}(\mathbb{R}^2,\{1,d\})>14.01$, 15.20 and 16.04 (Theorem 1), and for measurable colourings it holds for every real d. The thresholds up to 14 are close to the best the method of Lemma 19 can give: in floating point, the program of Lemma 19 gives k colours from about $d=5.797$, 10.42, 18.81, 35.53, 73.36, 179.5, 652.6 and 25 590 on, for $k=7,\dots,14$. Without spindles it gives $7,\dots,13$ colours from about 5.800, 10.67, 20.04, 40.04, 90.24, 261.7 and 1544 on and never 14; without triangles and spindles, from about 6.503, 12.03, 23.87, 52.85, 139.9, 599.2 and, for $k=13$, from somewhere between 50 000 and 54 300. For a single d the program with triangles is below $1/6$, for $1<d<5.8$, only for d between about 1.716 and 2.045 and from about

5.55 on. As $d \to \infty$ the value of the program without cliques tends to $\mu_0^2/(1+\mu_0)^2 = 1/12.13\ldots$, where $\mu_0 = -\min J_0 = 0.40275\ldots$: the square of its value $\mu_0/(1+\mu_0) = 0.287119$ for one distance [22, §3.2] (Remark after Lemma 19). With triangles and spindles its floating-point value at $D = 10^7$ is $0.075778 = 1/13.20$, close to the product $0.075362 = 1/13.27$ of 0.287119 and the value 0.262476 of the one-distance program with our triangles and spindles (with triangles alone the values are $1/12.80$, at $D = 10^8$, and 0.271823). So, in floating point, these configurations give no more than 14 colours for two distances; Theorem 9 gives more for very large d.

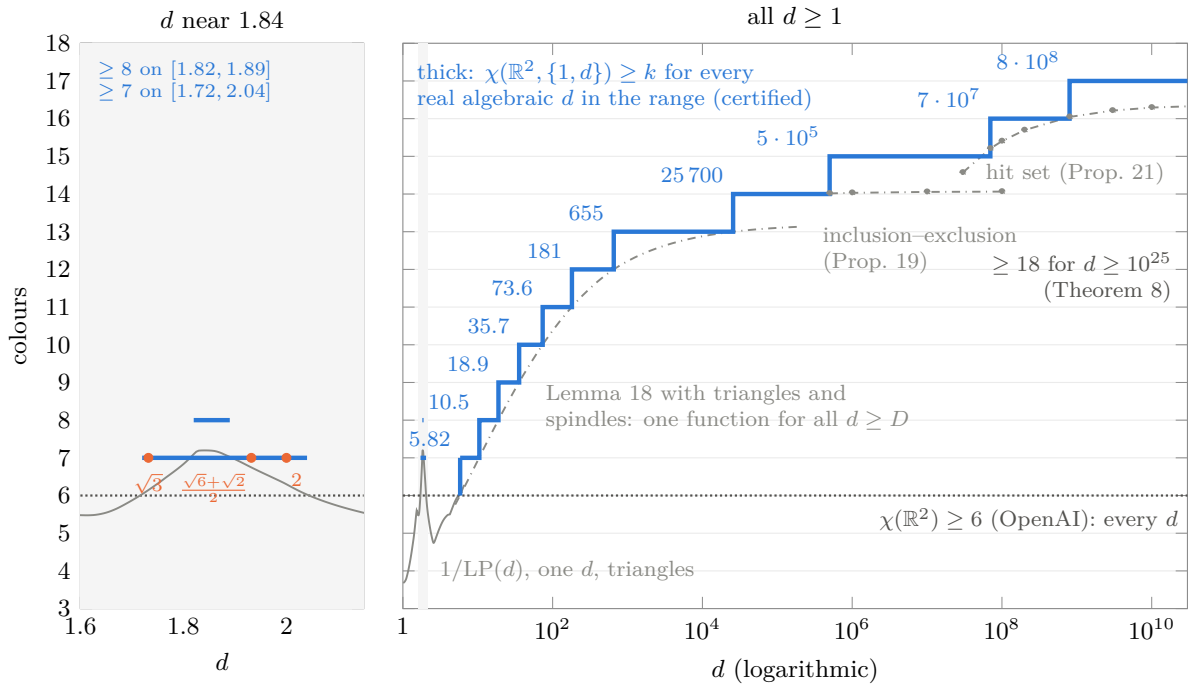


FIGURE 1. Lower bounds for $\chi(\mathbb{R}^2, \{1, d\})$, $d \geq 1$; the same hold for $1/d$. Thick blue: Theorem 5, for every real algebraic d in the range. Orange: the pairs $\{1, \sqrt{3}\}$, $\{1, 2\}$ and $\{1, (\sqrt{6}+\sqrt{2})/2\}$ of Corollary 4. Grey, solid: $1/\mathrm{LP}(d)$, where $\mathrm{LP}(d)$ is the value, in floating point (not a proof), of the linear program for the single distance d with unit-triangle cliques ($d \leq 6.6$); grey, dash-dotted: the same for the program of Lemma 19 with triangles and spindles, whose value at D serves every $d \geq D$. A certified bound below $1/(k-1)$ gives k colours, so each blue step lies less than one colour above the curve it follows. Grey dots on dash-dotted lines: the same for the programs of Propositions 20 and 22 (the latter with $L_B = 17.764$), at a few values of D. Dotted: $\chi(\mathbb{R}^2) \geq 6$ [23], which holds for every d. Left: the shaded range $1.6 \leq d \leq 2.15$ on a linear scale.

Many distances.

Corollary 6. *For every $k \geq 1$ there is a set D of k positive integers with $\chi(\mathbb{R}^d, D) \geq \chi(\mathbb{R}^2, D) \geq m_1(\mathbb{R}^2)^{-k} > 4.0485^k$ for every $d \geq 2$. Moreover*

$$\sup\{\chi_{\mathrm{f}}(\mathbb{R}^2, D) : D \subseteq F \cap (0,\infty),\ |D| = k\} = m_1(\mathbb{R}^2)^{-k},$$

where F is the field of real algebraic numbers.

Erdős [15] defined $L_2(r)$ as the largest chromatic number of the plane with r forbidden distances and asked: "Does $L_2(r)$ increase polynomially or exponentially? I expect that $L_2(r) < r^c$ for some absolute constant c." The question is Problem #706 on Bloom's list [14], where it is stated for finite point sets, which gives the same $L_2(r)$ by the de Bruijn–Erdős theorem; Croft, Falconer and Guy ask whether the growth is exponential [5, Problem G11] (as quoted by Bukh [2]), and the best lower bound in the plane was of order $k\sqrt{\log k}$ (see [2, 21]).

Corollary 7 (Erdős Problem #706)**.** *For every $r \geq 1$ there are positive integers $a_1 < \cdots < a_r$ and a finite set $P \subset \mathbb{R}^2$ such that the graph on P in which two points are joined when their distance is one of $a_1, \ldots, a_r$ has chromatic number greater than 4.0485^r. Consequently*

$$4.0485^r < L_2(r) \leq 7^r,$$

for every $r \geq 1$, both for $L_2(r)$ defined through the whole plane and through finite point sets; in particular $L_2(r)$ is not bounded by any polynomial in r.

Proof. Let $D = \{a_1, \ldots, a_r\}$ be the set of Corollary 6 for $k = r$, so $\chi(\mathbb{R}^2, D) > 4.0485^r$. Colouring the plane by a product of r hexagonal 7-colourings, the i-th scaled so that it avoids the distance a_i, gives $\chi(\mathbb{R}^2, D) \leq 7^r$; so $\chi(\mathbb{R}^2, D)$ is a finite integer $c > 4.0485^r$. By the de Bruijn–Erdős theorem [8], the graph on $\mathbb{R}^2$ with edge distances in D has a finite subgraph that is not $(c-1)$-colourable; let P be its vertex set. The upper bound $L_2(r) \leq 7^r$ holds for every r-element set of distances by the same product colouring, and a finite point set needs no more colours than the whole plane. □

So $L_2(r)$ grows exponentially, contrary to Erdős's expectation. Bukh [2] proved exponential growth for measurable colourings and remarked that it is "very likely that the usual chromatic number does not share this behavior"; Corollary 6 shows that it does, for integer distances. Davies [7] has just proved exponential growth for arbitrary colourings of $\mathbb{R}^d$, $d \geq 4$, with a different transfer (see the Remarks); Corollary 6 gives it in every dimension $d \geq 2$ (for $d \geq 4$ Davies's base $1/m_1(\mathbb{R}^d)$ is larger, for distances with rational squares). Like all our lower bounds, it rests on [23, Thm. 2.3]. Bukh also conjectured that $\chi(\mathbb{R}^d, D)$ is bounded independently of D when the elements of D are algebraically independent [2]; Theorem 12 says nothing about such D.

The same linear programs give explicit sets (Table 2):

Corollary 8. *$\chi(\mathbb{R}^2, \{1, 40, 1600\}) \geq 22$, $\chi(\mathbb{R}^2, \{1, 10, 100, 1000\}) \geq 26$, $\chi(\mathbb{R}^2, \{1, 100, 10^4\}) \geq 27$ and $\chi(\mathbb{R}^2, \{1, 30, 900, 27000\}) \geq 48$.*

These numbers are far below $m_1(\mathbb{R}^2)^{-k}$, which is approached only by very spread distances, but the sets are explicit and the bounds are certified.

Two scales. The lower bounds so far come from densities alone, so for two distances they never exceed the fractional chromatic number, which is below 19.01 for every pair $\{1, N\}$ (Corollary 10). Using OpenAI's theorem itself at one scale and densities at the other gives more.

Theorem 9. *Let D_1 and D_2 be finite sets of positive real algebraic numbers, let c be a positive integer with $c \leq \chi(\mathbb{R}^2, D_1)$, and let W be a finite graph drawn in the plane with all edge lengths in D_2. There is N_0 such that*

$$\chi(\mathbb{R}^2, D_1 \cup ND_2) \;\geq\; \chi_c(W) \;\geq\; c\,\chi_{\mathrm{f}}(W)$$

for every real algebraic number $N \geq N_0$. Consequently, with $c = \chi(\mathbb{R}^2, D_1)$,

$$\chi(\mathbb{R}^2, D_1 \cup ND_2) \;\geq\; \chi_c(\mathbb{R}^2, D_2) \;\geq\; \frac{c}{m_{D_2}(\mathbb{R}^2)}$$

for all large real algebraic N.

Here a *c-fold colouring* of a graph with k colours gives every vertex a set of c of the k colours, with disjoint sets at adjacent vertices, and the c-fold chromatic number χ_c is the least such k (Stahl's c-th chromatic number; see [18] for the plane). So $\chi_1 = \chi$, $\chi_c \geq c\,\chi_{\mathrm{f}}$ for finite graphs, and $\chi_c(\mathbb{R}^2, D) = \chi_c(G(\mathbb{C}, D))$.

At the scale of D_1, every disc of a fixed radius meets at least c classes of the measurable colouring given by Theorem 12, each in a set of measure bounded below, because a finite graph that needs c colours fits in the disc; at the scale of ND_2 these classes behave like a c-fold colouring. For graphs, the analogue is Geller and Stahl's formula $\chi(G[H]) = \chi_{\chi(H)}(G)$ for the lexicographic product [17], and Theorem 9 is the lower half of a geometric version; whether $\chi(\mathbb{R}^2, D_1 \cup ND_2) \to \chi_{\chi(\mathbb{R}^2, D_1)}(\mathbb{R}^2, D_2)$ as $N \to \infty$ is open (products of colourings give only

$\chi(\mathbb{R}^2, D_1)\,\chi(\mathbb{R}^2, D_2)$). The proof is in Section 5. For $c = 6$ no explicit finite graph is known, so N_0 is not explicit; with explicit inputs it is (the remark on explicit pairs in Section 5).

Corollary 10. (1) $\chi(\mathbb{R}^2, \{1, N\}) \geq 25$ *for every sufficiently large real algebraic number* N, *whereas* $\chi_{\mathrm{f}}(\mathbb{R}^2, \{1, N\}) \leq m_1(\mathbb{R}^2)^{-2} < 19.01$ *for every* $N > 0$.

(2) *There are integers* $1 < d_2 < d_3 < d_4$ *with* $\chi(\mathbb{R}^2, \{1, d_2\}) \geq 25$, $\chi(\mathbb{R}^2, \{1, d_2, d_3\}) \geq 102$ *and* $\chi(\mathbb{R}^2, \{1, d_2, d_3, d_4\}) \geq 413$, *and for every* $r \geq 1$ *there is a set* D *of* r *positive integers with* $\chi(\mathbb{R}^d, D) \geq \chi(\mathbb{R}^2, D) \geq 6 \cdot 4.0486^{r-1}$ *for every* $d \geq 2$. *Hence* $L_2(r) > 1.48 \cdot 4.0486^r$.

So for two distances the chromatic number can exceed the fractional chromatic number by a factor above $25/19.01 > 1.3$. For a single distance, OpenAI's theorem and Corollary 2 give the factor $6/4.36 > 1.37$. The bound $6/m_1(\mathbb{R}^2)$ that Theorem 9 gives for $\{1, N\}$ lies between 24.29 and 26.16, while products of hexagonal colourings give $\chi(\mathbb{R}^2, \{1, N\}) \leq 49$. The bound 25 needs only some finite unit-distance graph W with $\chi_{\mathrm{f}}(W) > 4$, such as the graph of Dúcz and Varga [12], so it does not depend on Theorem 13 or on [1]; the constants 102 and 413 do (from $\chi_{\mathrm{f}}(W) > 4.0007$ alone one gets 25, 101 and 405).

None of these results comes with an explicit finite graph: as in [23], the finite graphs exist by compactness.

2. Notation

Identify the plane with $\mathbb{C}$. Let $F = \overline{\mathbb{Q}} \cap \mathbb{R}$, let $E = F(i)$ be the plane of algebraic points and $K = \{u \in E : |u| = 1\}$ its group of rotations, as in [23, Def. 2.1]. For $D \subseteq (0, \infty)$ and $X \subseteq \mathbb{C}$ let $G(X, D)$ be the graph on X in which two points are adjacent when their distance lies in D. A set of real algebraic numbers is countable, and for $d \in F$ the points of E at distance d from 0 are exactly those of dK.

Fractional chromatic numbers. For a graph G with countably many vertices, $\chi_{\mathrm{f}}(G)$ is the least t such that some Borel probability measure on the closed set of independent sets of G in $\{0,1\}^{V(G)}$ gives every vertex probability at least $1/t$ of lying in the random set. For finite G this is the usual fractional chromatic number, and by weak-$*$ compactness $\chi_{\mathrm{f}}(G)$ is the supremum over the finite subgraphs of G. We write $\chi_{\mathrm{f},0}(\mathbb{R}^2, D)$ for the supremum of $\chi_{\mathrm{f}}(H)$ over finite graphs H drawn in the plane with all edge lengths in D, the *finitary* fractional chromatic number of [20] when $D = \{1\}$, and $\chi_{\mathrm{f}}(\mathbb{R}^2, D)$ for the infimum of $\sum_I w(I)$ over finitely supported weightings $w \geq 0$ of independent sets of $G(\mathbb{C}, D)$ with $\sum_{I \ni x} w(I) \geq 1$ for every point x. Clearly $\chi_{\mathrm{f},0} \leq \chi_{\mathrm{f}}$. Since F is real closed, Tarski's transfer principle lets every finite graph drawn with edge lengths in $D \subseteq F$ be redrawn with vertices in E; so, for $D \subseteq F$,

$$\chi_{\mathrm{f},0}(\mathbb{R}^2, D) = \chi_{\mathrm{f}}(G(E, D)) \quad \text{and} \quad \chi(\mathbb{R}^2, D) = \chi(G(E, D)).$$

Densities. The upper density of a measurable $A \subseteq \mathbb{C}$ is

$$\overline{\delta}(A) = \limsup_{R \to \infty} \frac{|A \cap B(0, R)|}{|B(0, R)|}.$$

A set *avoids* D if no two of its points are at a distance in D, and $m_D(\mathbb{R}^2)$ is the supremum of $\overline{\delta}(A)$ over measurable sets A avoiding D. For finite D it is also the supremum of the densities of periodic sets avoiding D (Lemma 14; the same proof works with squares in place of discs), so it does not matter whether densities are taken over discs, as here, or over squares, as in [22, 2]. A measurable set A *weakly avoids* D if

$$\int_{B(0,R)} \int_{S^1} 1_A(x)\, 1_A(x + du)\, d\sigma(u)\, dx = 0 \qquad \text{for all } d \in D,\ R > 0,$$

where σ is the normalized arc length. A *weak measurable k-colouring avoiding* D is a measurable map $c : \mathbb{C} \to \{1, \dots, k\}$ whose classes weakly avoid D.

3. The transfers

From [23] we use Lemma 4.1 (the continuous vectors of a measure-preserving action of E form L^2 of a factor, and the translations extend to a strongly continuous action T_x, $x \in \mathbb{R}^2$, on it), Theorem 2.3 (every K-invariant Borel probability measure on the character group of the discrete

group E that gives measure zero to the continuous characters is Haar measure), the measurable sampling step in the proof of Proposition 4.2, and Lemma 4.3. Lemma 4.1 and Theorem 2.3 involve no distance; Lemma 4.3 and the sampling step concern the unit circle, and we redo them at radius d (Lemma 11 and the proofs below).

Lemma 11. *If a measurable set A weakly avoids D, then the set A^* of points of density one of A avoids D, and $|A \setminus A^*| = 0$.*

Proof. This is [23, Lemma 4.3] at radius d. Let $x,\ y = x + v \in A^*$ with $|v| = d \in D$. For small $h > 0$ choose x' uniformly in $B(x, h)$ and, independently, $v' = du'$ with u' uniform on the arc $\{u \in S^1 : |du - v| < h\}$. Then $P(x' \notin A) \to 0$. The point $x' + v'$ is a mixture of uniform points of discs of radius h, so it has a density at most $1/(\pi h^2)$, and it lies in $B(y, 2h)$; hence $P(x' + v' \notin A) \le |B(y, 2h) \setminus A|/(\pi h^2) \to 0$. For small h both points lie in A with positive probability, while the law of (x', u') is absolutely continuous with respect to $dx\, d\sigma(u)$. This contradicts weak avoidance. The second statement is Lebesgue's density theorem. □

Theorem 12. *Let $D \subseteq F$ be a set of positive real algebraic numbers and $k \ge 1$. If $G(E, D)$ has a proper k-colouring, the plane has a weak measurable k-colouring avoiding D; conversely, such a colouring gives a proper k-colouring of $G(\mathbb{C}, D)$. In particular $\chi(\mathbb{R}^2, D) \ge 1/m_D(\mathbb{R}^2)$.*

Proof. This is the proof of [23, Thm. 1.3] with the unit circle replaced by the circles of radii $d \in D$.

(1) Let $\Omega_D \subseteq \{1, \dots, k\}^E$ be the closed set of proper colourings of $G(E, D)$, nonempty by hypothesis. Translations by E and rotations by K preserve it. The group $E \rtimes K$ is amenable, so averages of a point mass along a Følner sequence have an invariant weak-$*$ limit μ.

(2) Let f_i be the indicator that the origin has colour i. Properness gives $\langle f_i, T_z f_i \rangle = 0$ for $z \in dK$, $d \in D$.

(3) By [23, Lemma 4.1], $f_i = p_i + b_i$ where p_i is the conditional expectation on the factor of continuous vectors; $0 \le p_i \le 1$, $\sum_i p_i = 1$, both parts are rotation-fixed, and the spectral measure of b_i is K-invariant and null on the continuous characters. By [23, Thm. 2.3] it is a multiple of Haar measure, so $\langle b_i, T_z b_i \rangle = 0$ for $z \in E \setminus \{0\}$; the cross terms vanish because the continuous subspace and its orthogonal complement are translation invariant. Hence $\langle p_i, T_z p_i \rangle = 0$ for $z \in dK$.

(4) The T_x act strongly continuously and dK is dense in the circle of radius d, so $\langle p_i, T_x p_i \rangle = 0$ whenever $|x| \in D$.

(5) The sampling step of [23, Prop. 4.2] gives jointly measurable fields $H_i(x, \omega)$ representing $T_x p_i$. Since D is countable, a single sample ω makes all the integrals

$$\int_{B(0,m)} \int_{S^1} H_i(x, \omega) H_i(x + du, \omega)\, d\sigma(u)\, dx \qquad (m \in \mathbb{N},\ d \in D,\ \text{all } i)$$

vanish and has $\sum_i H_i(x, \omega) = 1$ for almost every x. Giving x the least i with $H_i(x, \omega) > 0$ defines the weak measurable colouring.

(6) Conversely, by Lemma 11 the density-one points of the classes form disjoint sets avoiding D whose union is conull. Every finite configuration has a translate inside that union, so every finite subgraph of $G(\mathbb{C}, D)$ is k-colourable, and the de Bruijn–Erdős theorem [8] gives a proper colouring. One of the k sets has upper density at least $1/k$, so $k \ge 1/m_D(\mathbb{R}^2)$. □

Formal verification. Theorem 12, in the form used in Section 5, is formally verified in Lean 4 (version 4.34.1, Mathlib commit `d13f23b7`) on top of OpenAI's formalization of their proof (`https://github.com/openai/math`, commit `fd4aeeb2`): for every set D of positive real algebraic numbers, a proper k-colouring of $G(\mathbb{C}, D)$ gives a measurable k-colouring whose classes weakly avoid every $d \in D$ (`OAI.Ext.proper_to_borel_weak_distances`), and k pairwise disjoint measurable sets, each avoiding D, whose union is conull (`OAI.Ext.proper_to_measurable_classes`); Lemma 11 at radius r is `OAI.Ext.weak_density_proper_radius`. The proofs use only the standard axioms (`propext`, `Classical.choice`, `Quot.sound`); the files are in `lean/`

`external/openai-five/distances/` of [27]. Not formalized: the converse of Theorem 12, the inequality $\chi \geq 1/m_D$, Theorem 13, the lemmas of Section 5 and the density bounds.

Theorem 13. *For every set $D \subseteq F$ of positive real algebraic numbers, $\chi_{\mathrm{f},0}(\mathbb{R}^2, D) \geq 1/m_D(\mathbb{R}^2)$.*

Proof. Let $t \geq \chi_{\mathrm{f},0}(\mathbb{R}^2, D) = \chi_{\mathrm{f}}(G(E, D))$ and let P be a Borel probability measure on the compact metrizable space Ω of independent sets of $G(E, D)$ with $P(z \in I) \geq 1/t$ for all $z \in E$. Averaging the push-forwards of P under $E \rtimes K$ along a Følner sequence and taking a weak-$*$ limit gives an invariant probability μ with $\mu(0 \in I) \geq 1/t$, since each push-forward has this property and the event is clopen. Let $f = 1[0 \in I]$. It is rotation-fixed, and $\langle f, T_z f\rangle = \mu(0 \in I,\ z \in I) = 0$ for $z \in dK$, $d \in D$, because I is independent. Steps 3 and 4 of the proof of Theorem 12 apply to the single function f, with $\sum_i p_i = 1$ (which is used only in Step 5) replaced by $\mathbb{E}_\mu p = \mathbb{E}_\mu f$: for $p = \mathbb{E}(f \mid \text{continuous factor})$ we get $0 \leq p \leq 1$, $\mathbb{E}_\mu p \geq 1/t$ and $\langle p, T_x p\rangle = 0$ whenever $|x| \in D$.

Take a jointly measurable $H : \mathbb{C} \times \Omega \to [0, 1]$ with $H(x, \cdot) = T_x p$ in $L^2(\mu)$ for almost every x (the sampling step of [23, Prop. 4.2]). For $m \in \mathbb{N}$ and $d \in D$,

$$\int_\Omega \int_{B(0,m)} \int_{S^1} H(x, \omega) H(x + du, \omega)\, d\sigma(u)\, dx\, d\mu(\omega) = \int_{B(0,m)} \int_{S^1} \langle T_x p, T_{x+du} p\rangle\, d\sigma(u)\, dx = 0,$$

so, D being countable, for μ-almost every ω all these inner integrals vanish. Let $X_n(\omega) = |B(0, n)|^{-1} \int_{B(0,n)} H(x, \omega)\, dx \in [0, 1]$. Since $\mathbb{E}_\mu H(x, \cdot) = \langle T_x p, 1\rangle = \langle p, 1\rangle \geq 1/t$ for almost every x ($T_x 1 = 1$ and T_x is unitary), $\mathbb{E}_\mu X_n \geq 1/t$, and the reverse Fatou lemma gives $\mathbb{E}_\mu \limsup_n X_n \geq 1/t$. Hence $\limsup_n X_n \geq 1/t$ on a set of positive measure, and we choose ω in it and in the conull set above. Let $A = \{x : H(x, \omega) > 0\}$. Then $1_A \geq H(\cdot, \omega)$, so $\overline{\delta}(A) \geq 1/t$; and $H(x, \omega) H(x + du, \omega) = 0$ for almost every (x, u) gives $1_A(x) 1_A(x + du) = 0$ almost everywhere, so A weakly avoids D. By Lemma 11, A^* avoids D and $\overline{\delta}(A^*) \geq 1/t$. Hence $m_D(\mathbb{R}^2) \geq 1/t$. □

4. The upper bound and the corollaries

Lemma 14. *Let D be finite, $M = \max D$ and $\varepsilon > 0$. There are $\eta > 0$ and a periodic set A' with period lattice $N\mathbb{Z}^2$, each period a finite union of squares, that avoids $D_\eta = D + [-\eta, \eta]$ and has density at least $m_D(\mathbb{R}^2) - \varepsilon$.*

Proof. Let A be measurable, avoiding D, with $\overline{\delta}(A) > m_D - \varepsilon/4$, and fix $L > 0$. For the radii R realising the upper density, with R large, the squares of the grid $L\mathbb{Z}^2$ meeting $B(0, R)$ (at most $\pi(R + \sqrt{2}L)^2/L^2$ of them) have average density of A above $m_D - \varepsilon/2$, so one of them, S, has $|A \cap S| \geq (m_D - \varepsilon/2)L^2$. By inner regularity there is a compact $C \subseteq A \cap S$ with $|C| \geq (m_D - 3\varepsilon/4)L^2$. Its distance set $\{|x - y| : x, y \in C\}$ is compact and misses the finite set D, so it misses $D_{2\eta}$ for some $\eta \in (0, 1)$. Let U be the union of the closed squares of a grid of squares of diameter less than $\eta/2$ that meet C. It lies in the $\eta/2$-neighbourhood of C, so it avoids D_η, and $|U| \geq |C|$. The set $A' = U + N\mathbb{Z}^2$ with $N = L + M + 2$ avoids D_η, since points of different copies are more than $M + 1$ apart, and has density $|U|/N^2 \geq m_D - \varepsilon$ once L is large. □

Proof of Theorem 1. By Theorem 13, $\chi_{\mathrm{f}}(\mathbb{R}^2, D) \geq \chi_{\mathrm{f},0}(\mathbb{R}^2, D) \geq 1/m_D$, and $\chi \geq \chi_{\mathrm{f},0}$ always. Let A' be as in Lemma 14, with density $\delta \geq m_D - \varepsilon$ and fundamental square $Q = [0, N)^2$. For $n \geq 1$ take the n^2 translates $A' + s$, $s \in (N/n)\mathbb{Z}^2 \cap Q$; each avoids D. A point x lies in $A' + s$ if and only if $x - s \in A'$, so the number of these translates containing x is the number of points of the grid $x - (N/n)\mathbb{Z}^2$ in one period of A'. One period is a finite union of squares, so its boundary meets $O(n)$ cells of that grid, uniformly in x, and the count is at least $\delta n^2 - O(n)$ for every x. Giving each translate the weight $1/(\delta n^2 - O(n))$ is a fractional colouring of $G(\mathbb{C}, D)$ with total weight tending to $1/\delta$. Hence $\chi_{\mathrm{f}}(\mathbb{R}^2, D) \leq 1/(m_D - \varepsilon)$ for every $\varepsilon > 0$. □

Proof of Corollaries 2–6. Corollary 2 is Theorem 1 with $D = \{1\}$, together with $0.229364 \leq m_1 \leq 0.2470$ [6, 1]; [1, §8] proves $m_1 \leq (w_T + \nu)/(1 - \mu)$ with $w_T = 0.24697262945998308$, $\nu = 10^{-5}$ and $\mu = 6 \cdot 10^{-5}$, which is less than 0.2469975.

Corollary 3. If X is a set of n points, its unit-distance graph G has $\chi_{\mathrm{f}}(G) \leq \chi_{\mathrm{f},0}(\mathbb{R}^2) = 1/m_1(\mathbb{R}^2)$, so $\alpha(G) \geq n/\chi_{\mathrm{f}}(G) \geq m_1(\mathbb{R}^2)\, n$. Conversely, [20, Thm. 2] states $\chi_{\mathrm{f},0}(\mathbb{R}^2) = 1/\alpha_1(\mathbb{R}^2)$, where $\alpha_1(\mathbb{R}^2)$ is the infimum of $\alpha(G)/|V(G)|$ over the unit-distance graphs G of finite point sets

in the plane, so $\inf_n f(n)/n = \alpha_1(\mathbb{R}^2) = m_1(\mathbb{R}^2)$ by Corollary 2. Placing two point sets far apart shows $f(a+b) \le f(a) + f(b)$, so by Fekete's lemma $f(n)/n$ converges to $\inf_n f(n)/n = m_1(\mathbb{R}^2)$.

Corollary 4. The finite bounds follow from Theorem 12 and the density bounds of Table 1. For the last statement, Bukh's Theorem 1 [2] gives $m_{\{1,d\}}(\mathbb{R}^2) \to m_1(\mathbb{R}^2)^2 \le 0.2470^2 < 1/16$ as $d \to \infty$; Theorem 12 then gives $\chi(\mathbb{R}^2, \{1, d\}) \ge 17$ for large algebraic d. Corollary 8 follows in the same way from Table 2.

Corollary 6. Bukh's Theorem 1 gives $m_D(\mathbb{R}^2) \to m_1(\mathbb{R}^2)^k$ for $D = \{1, t, \dots, t^{k-1}\}$ as $t \to \infty$. For integers t, Theorem 12 gives $\chi(\mathbb{R}^2, D) \ge 1/m_D(\mathbb{R}^2)$; since $\chi(\mathbb{R}^2, D)$ is an integer and $1/m_D(\mathbb{R}^2) \to m_1(\mathbb{R}^2)^{-k}$, we get $\chi(\mathbb{R}^2, D) \ge m_1(\mathbb{R}^2)^{-k}$ for large t. Restricting a colouring of $\mathbb{R}^d$ to a plane gives $\chi(\mathbb{R}^d, D) \ge \chi(\mathbb{R}^2, D)$. For the supremum, Theorem 1 gives $\chi_{\mathrm{f}}(\mathbb{R}^2, D) = 1/m_D(\mathbb{R}^2)$ for finite $D \subseteq F$; Bukh proves $m_D(\mathbb{R}^2) \ge m_1(\mathbb{R}^2)^k$ whenever $|D| = k$ [2], and the sets $\{1, t, \dots, t^{k-1}\}$ show that the bound $m_1(\mathbb{R}^2)^{-k}$ is approached. (By [22, Thm. 5.1], $m_D(\mathbb{R}^2) \le 2^{-k}$ as soon as consecutive ratios exceed some $r(k)$ that their proof makes effective, which gives $\chi(\mathbb{R}^2, D) \ge 2^k$ for such D made of algebraic numbers.) □

5. Two scales

In this section $B(x, R)$ is the open disc, σ_d is the normalized arc length on the circle of radius d about 0, and $(f * \sigma_d)(x) = \int_{S^1} f(x + du)\, d\sigma(u)$. For a bounded measurable $f \ge 0$ and $d > 0$ let

$$P_d(f) = \limsup_{L \to \infty} \frac{1}{|B(0, L)|} \int_{B(0,L)} f\,(f * \sigma_d)\, dx,$$

the upper density of pairs at distance d, and $P_d(S) = P_d(1_S)$; a set that avoids d has $P_d(S) = 0$.

Lemma 15. *Let G be a finite graph drawn in the plane with all edge lengths in D_1 and $\chi(G) \ge c$, with n vertices, all in $B(0, R_0)$. Let $A_1, \dots, A_k$ be measurable sets, each avoiding D_1, whose union is conull. Then $k \ge c$, and for every $x \in \mathbb{C}$ at least c indices i satisfy*

$$|A_i \cap B(x, 2R_0)| \;\ge\; \frac{|B(x, 2R_0)|}{4nk}.$$

Proof. Let $\Delta = B(x, 2R_0)$, let J be a set of $\min(c-1, k)$ indices i with the largest values of $|A_i \cap \Delta|$, and let $U = \Delta \setminus \bigcup_{j \in J} A_j$. For $t \in B(x, R_0)$ the vertices of $G + t$ lie in Δ. If none of them were in U, each would lie in some A_j with $j \in J$; choosing one such j for every vertex would give a proper colouring of G with at most $c-1$ colours, since two adjacent vertices are at a distance in D_1 and no A_j contains two such points. Hence every $t \in B(x, R_0)$ lies in $\bigcup_{v \in V(G)} (U - v)$, and $|U| \ge |B(0, R_0)|/n = |\Delta|/(4n) > 0$. The union of the sets A_i is conull, so $|U| \le \sum_{i \notin J} |A_i \cap \Delta|$. Therefore some index lies outside J, which forces $k \ge c$ and $|J| = c - 1$, and the largest of the $k - c + 1$ terms of this sum is at least $|\Delta|/(4n(k-c+1)) \ge |\Delta|/(4nk)$. By the choice of J, the $c-1$ indices in J satisfy the same bound. □

Lemma 16. *Let A be a measurable set avoiding the distance $d > 0$, let $R > 0$ and $\rho(x) = |A \cap B(x, R)|/(\pi R^2)$. For every $L > 0$,*

$$\int_{B(0,L)} \rho\,(\rho * \sigma_d)\, dx \;\le\; \sqrt{\frac{R}{\pi d}}\, |A \cap B(0, L + d + R)|, \qquad \text{so} \qquad P_d(\rho) \le \sqrt{\frac{R}{\pi d}}\, \overline{\delta}(A).$$

Proof. Let h be the indicator of $A \cap B(0, L + d + R)$, $\varphi = 1_{B(0,R)}/(\pi R^2)$ and $F = h * \varphi$. For $|x| < L$ and $|u| = 1$ the discs $B(x, R)$ and $B(x + du, R)$ lie in $B(0, L + d + R)$, so $F(x) = \rho(x)$ and $F(x + du) = \rho(x + du)$. Since $F \ge 0$, Plancherel's theorem gives

$$\int_{B(0,L)} \rho\,(\rho * \sigma_d)\, dx \;\le\; \int_{\mathbb{R}^2} F\,(F * \sigma_d)\, dx \;=\; \int_{\mathbb{R}^2} |\hat{h}(\xi)|^2\, \Lambda(2\pi R|\xi|)^2\, J_0(2\pi d|\xi|)\, d\xi,$$

where $\hat{f}(\xi) = \int f(x) e^{-2\pi i \langle x, \xi \rangle} dx$, $\hat{\sigma}_d(\xi) = J_0(2\pi d|\xi|)$, $\hat{\varphi}(\xi) = \Lambda(2\pi R|\xi|)$ and $\Lambda(s) = 2J_1(s)/s$. As A avoids d, $h(x)h(x + du) = 0$ for all x and u, so $0 = \int h\,(h * \sigma_d)\, dx = \int |\hat{h}(\xi)|^2 J_0(2\pi d|\xi|)\, d\xi$, and the right-hand side equals $\int |\hat{h}(\xi)|^2\, (\Lambda(s)^2 - 1) J_0(qs)\, d\xi$ with $s = 2\pi R|\xi|$ and $q = d/R$.

We bound $(1 - \Lambda(s)^2)\, |J_0(qs)|$. First, $|\Lambda| \le 1$ because $\hat{\varphi}$ is the Fourier transform of a probability density. For $0 \le s \le 2\sqrt{2}$ the series $\Lambda(s) = \sum_{m \ge 0} (-1)^m (s/2)^{2m}/(m!\,(m+1)!)$ alternates with

nonincreasing terms, so $\Lambda(s) \geq 1 - s^2/8 \geq 0$ and $1 - \Lambda(s)^2 \leq 2(1 - \Lambda(s)) \leq s^2/4$; for $s > 2\sqrt{2}$, $1 - \Lambda(s)^2 \leq 1 < s^2/4$. Second, $|J_0(y)| \leq \min(1, \sqrt{2/(\pi y)})$ for $y > 0$ (Section 6). For $s \leq 2$ the product is at most $\frac{s^2}{4}\sqrt{2/(\pi q s)} = \frac{1}{4}s^{3/2}\sqrt{2/(\pi q)} \leq (\pi q)^{-1/2}$, and for $s \geq 2$ at most $\sqrt{2/(\pi q s)} \leq (\pi q)^{-1/2}$. Hence the integral is at most $(\pi q)^{-1/2} \int |\hat{h}|^2 = \sqrt{R/(\pi d)}\, |A \cap B(0, L+d+R)|$. Dividing by $|B(0,L)|$ and letting $L \to \infty$ gives the second inequality. □

Lemma 17. *Let W be a finite graph drawn in the plane, let $c, k \geq 1$ be integers with $k < \chi_c(W)$, and let $S_1, \dots, S_k$ be measurable sets such that every point of the plane lies in at least c of them. Then for every $N > 0$*

$$\sum_{i=1}^{k} \sum_{ww' \in E(W)} P_{N|w-w'|}(S_i) \;\geq\; 1.$$

Proof. For $x \in \mathbb{C}$ and $\theta \in S^1$ let $T(w) = \{i : x + N\theta w \in S_i\}$, a set of at least c indices. If $T(w) \cap T(w') = \emptyset$ for every edge ww', then c-element subsets of the sets $T(w)$ form a c-fold colouring of W with $k < \chi_c(W)$ colours, which is impossible. Hence

$$\sum_{i} \sum_{ww' \in E(W)} 1_{S_i}(x + N\theta w)\, 1_{S_i}(x + N\theta w') \;\geq\; 1$$

for all x and θ. Integrate over $x \in B(0,L)$ and over θ (normalized Haar measure on S^1), and let $r = \max_w |w|$. For an edge ww', the substitution $y = x + N\theta w$ and the fact that $\theta(w' - w)$ is uniformly distributed on the circle of radius $|w' - w|$ give

$$\int_{S^1} \int_{B(0,L)} 1_{S_i}(x + N\theta w)\, 1_{S_i}(x + N\theta w')\, dx\, d\theta \;\leq\; \int_{B(0,L+Nr)} 1_{S_i}\, (1_{S_i} * \sigma_{N|w-w'|})\, dy.$$

Divide by $|B(0,L)|$ and take $\limsup_{L\to\infty}$; since $|B(0,L+Nr)|/|B(0,L)| \to 1$ and the limit superior of a finite sum is at most the sum of the limits superior, the claim follows. □

Proof of Theorem 9. The graph $G(\mathbb{C}, D_1)$ has chromatic number at least c, so by the de Bruijn–Erdős theorem [8] some finite graph G drawn in the plane with edge lengths in D_1 has $\chi(G) \geq c$; translate it so that its n vertices lie in $B(0, R_0)$, and put $R = 2R_0$. If $\chi_c(W) = 1$ there is nothing to prove. Otherwise let K be an integer with $1 \leq K < \chi_c(W)$ (any such K works, so a lower bound for $\chi_c(W)$ suffices), put $\varepsilon = 1/(4nK)$, and let $N > 0$ be a real algebraic number with

$$K\, |E(W)|\, \varepsilon^{-2} \sqrt{R/(\pi N \min D_2)} \;<\; 1,$$

which holds for all large N. We show that $\chi(\mathbb{R}^2, D_1 \cup ND_2) > K$. Suppose instead that $G(\mathbb{C}, D_1 \cup ND_2)$ has a proper k-colouring with $k \leq K$. By Theorem 12 and step 6 of its proof, there are pairwise disjoint measurable sets $A_1, \dots, A_k$, each avoiding $D_1 \cup ND_2$, whose union is conull. Let $\rho_i(x) = |A_i \cap B(x,R)|/(\pi R^2)$, a continuous function, and $S_i = \{x : \rho_i(x) \geq \varepsilon\}$. By Lemma 15, since $1/(4nk) \geq \varepsilon$, every point lies in at least c of the sets S_i. Since $1_{S_i} \leq \rho_i/\varepsilon$ and A_i avoids Nd for $d \in D_2$, Lemma 16 gives $P_{Nd}(S_i) \leq \varepsilon^{-2} P_{Nd}(\rho_i) \leq \varepsilon^{-2}\sqrt{R/(\pi N d)}$. As $k \leq K < \chi_c(W)$, Lemma 17 gives

$$1 \;\leq\; k\, |E(W)|\, \varepsilon^{-2} \sqrt{R/(\pi N \min D_2)} \;<\; 1,$$

a contradiction. Taking $K = \chi_c(W) - 1$ gives the first inequality, and $\chi_c(W) \geq c\, \chi_{\mathrm{f}}(W)$ holds for every finite graph.

For the second statement, $c = \chi(\mathbb{R}^2, D_1)$ is finite (a product of hexagonal colourings, one for each distance, uses $7^{|D_1|}$ colours), and so is $\chi_c(\mathbb{R}^2, D_2) \leq c\, 7^{|D_2|}$ (replace each colour by c new ones). A graph has a c-fold colouring with k colours if and only if it has a homomorphism into the finite Kneser graph of c-subsets of $\{1, \dots, k\}$, so by compactness, as in the de Bruijn–Erdős theorem, some finite graph W drawn with edge lengths in D_2 has $\chi_c(W) = \chi_c(\mathbb{R}^2, D_2)$; apply the first statement to it. Finally $\chi_c(\mathbb{R}^2, D_2) \geq c \sup_W \chi_{\mathrm{f}}(W) = c\, \chi_{\mathrm{f},0}(\mathbb{R}^2, D_2) \geq c/m_{D_2}(\mathbb{R}^2)$ by Theorem 13. □

Proof of Corollary 10. By Corollary 2, $\chi_{\mathrm{f},0}(\mathbb{R}^2) = 1/m_1(\mathbb{R}^2) > 1/0.2469975 > 4.0486$, so some finite unit-distance graph W has $\chi_{\mathrm{f}}(W) > 4.0486$. (1) Theorem 9 with $D_1 = D_2 = \{1\}$, $c = 6$ [23] and this W gives $\chi(\mathbb{R}^2, \{1, N\}) \geq \chi_6(W) \geq 6\,\chi_{\mathrm{f}}(W) > 24.29$ for large N. For every $N > 0$, $m_{\{1,N\}}(\mathbb{R}^2) \geq m_1(\mathbb{R}^2)^2$ (Bukh [2]; intersect a periodic set avoiding 1 with a suitable translate of a dilate by N of another), and the proof of the upper bound in Theorem 1 does not use algebraicity, so $\chi_{\mathrm{f}}(\mathbb{R}^2, \{1, N\}) \leq m_1(\mathbb{R}^2)^{-2} \leq 0.229364^{-2} < 19.01$. (2) Let $c_1 = 6$ and $D^{(1)} = \{1\}$. Given a finite set $D^{(j)}$ of integers with $\chi(\mathbb{R}^2, D^{(j)}) \geq c_j$, Theorem 9 with $D_1 = D^{(j)}$, $D_2 = \{1\}$ and this W gives an integer $d_{j+1} > \max D^{(j)}$ with $\chi(\mathbb{R}^2, D^{(j)} \cup \{d_{j+1}\}) > 4.0486\,c_j$; let c_{j+1} be the least integer above $4.0486\,c_j$. Then $c_2 = 25$, $c_3 = 102$, $c_4 = 413$ and $c_r \geq 6 \cdot 4.0486^{r-1}$, which exceeds $1.48 \cdot 4.0486^r$. Restricting colourings of $\mathbb{R}^d$ to a plane gives the bounds for $d \geq 2$. □

Remark (Explicit pairs). The proof is effective when G and W are explicit. Take $c = 5$, and for G de Grey's unit-distance graph with 1581 vertices, 7877 edges and no proper 4-colouring [9], whose vertices lie within distance 2.27 of one of them (we computed 2.2641 from exact coordinates). For W take the Moser spindle, with 7 vertices, 11 edges and independence number 2, so that $\chi_5(W) \geq 5 \cdot 7/2$, that is $\chi_5(W) \geq 18$. With $K = 17$, $n = 1581$, $\varepsilon = 1/(4nK) = 1/107508$ and $R = 4.54$, the condition $K\,|E(W)|\,\varepsilon^{-2}\sqrt{R/(\pi N)} < 1$ of the proof holds as soon as $N > (17 \cdot 11 \cdot 107508^2)^2 \cdot 4.54/\pi \approx 6.75 \cdot 10^{24}$. Hence $\chi(\mathbb{R}^2, \{1, N\}) \geq 18$ for every real algebraic $N \geq 10^{25}$; density bounds give 17 for every real algebraic $N \geq 8 \cdot 10^8$ (Theorem 5). For W one can also take the 607-vertex unit-distance graph of Bellitto, Pêcher and Sédillot, whose weighting has weighted independence ratio 512933/1999983 [3, §4.4], so that $\chi_{\mathrm{f}}(W) \geq 1999983/512933 > 3.8991$ (cf. [3, Thm. 2]) and $\chi_5(W) \geq 20$; only $\chi_{\mathrm{f}}(W) > 3.8$ is needed. With $K = 19$, $\varepsilon = 1/120156$ and the trivial bound $|E(W)| \leq \binom{607}{2} = 183921$, the condition holds for $N > 3.68 \cdot 10^{33}$, so $\chi(\mathbb{R}^2, \{1, N\}) \geq 20$ for every real algebraic $N \geq 10^{34}$, on the strength of their computation. These bounds avoid OpenAI's theorem $\chi(\mathbb{R}^2) \geq 6$ but still rest on Theorem 12, that is, on OpenAI's rigidity theorem.

6. Linear-programming certificates

Proposition 18 (Oliveira Filho–Vallentin [22], with configuration inequalities)**.** *Let $D = \{d_1, \dots, d_\ell\}$, let $W_1, \dots, W_r$ be finite planar point sets and $p_1, \dots, p_r$ points, and let α_c be the largest number of points of W_c no two of which are at a distance in D (so $\alpha_c = 1$ when all pairwise distances of W_c lie in D). Suppose $z_0, \dots, z_\ell \in \mathbb{R}$ and $y_1, \dots, y_r \geq 0$ satisfy*

$$g(t) = z_0 + \sum_k z_k J_0(d_k t) + \sum_c y_c \sum_{w \in W_c} J_0(|w - p_c|\,t) \;\geq\; 0 \qquad (t \geq 0),$$

with $s = z_0 + \sum_k z_k + \sum_c y_c |W_c| > 0$. Then $m_D(\mathbb{R}^2) \leq (z_0 + \sum_c \alpha_c y_c)/s$.

Proof. By Lemma 14 it suffices to consider a periodic measurable set A of density $\delta > 0$ avoiding D. Its autocorrelation $\varphi(v) = \delta(A \cap (A - v))$ is positive definite, with spectral measure ν on the dual lattice, $\nu(\{0\}) = \delta^2$, total mass δ, so that $\varphi(v) = \int e^{2\pi i \langle \xi, v \rangle}\, d\nu(\xi)$; its radial average is $f(r) = \int J_0(2\pi r|\xi|)\, d\nu(\xi)$, with $f(0) = \delta$ and $f(d_k) = 0$. For a finite set W, its number α as above, and any p, every point x of A lies in at most α of the sets $\{x \in A : x + w - p \in A\}$, $w \in W$, because the points $x + w - p$ that lie in A have no two at a distance in D; so $\sum_w \varphi(w - p) \leq \alpha\delta$, and averaging over rotations of W about p gives $\sum_w f(|w - p|) \leq \alpha\delta$. Integrating $g(2\pi|\xi|) \geq 0$ against ν, $\delta^2 g(0) \leq \int g(2\pi|\xi|)\, d\nu = z_0\delta + \sum_c y_c \sum_w f(|w - p_c|) \leq (z_0 + \sum_c \alpha_c y_c)\delta$, and $g(0) = s$. Without the terms in y_c this is [22, Thm. 1.1]. The terms with $\alpha_c = 1$ are in the spirit of [22, §3.1], which uses simplices (we also use a rhombus), and the inequality for a finite unit-distance graph with independence number α is [19, Lemma 3.2], where Keleti, Matolcsi, Oliveira Filho and Ruzsa use copies of the Moser spindle ($\alpha = 2$) to bound $m_1(\mathbb{R}^2)$; for $D \ni 1$ the number α_c of a unit-distance graph is at most its independence number. □

We solve the linear program in floating point, round the coefficients to rationals and verify $g \geq 0$ rigorously. Three programs check every certificate.

- `certify_lp2.py` uses Arb ball arithmetic, with distances and radii enclosed from their exact expressions. On $[0, T]$ it walks with adaptive steps: at a point t it encloses $g(t)$

and moves on by about $g(t)/L$, where $L = \sum |\text{coefficient}| \cdot \text{radius}/\sqrt{2}$ bounds $|g'|$ because $|J_1(x)| \le 1/\sqrt{2}$ for real x [11, 10.14.1]. On $[T, \infty)$ it uses $|J_0(x)| \le \sqrt{2/(\pi x)}$ for $x > 0$: $u = \sqrt{x} J_0(x)$ solves $u'' + (1 + 1/(4x^2))u = 0$, so $u^2 + u'^2/(1 + 1/(4x^2))$ is nondecreasing (Sonin–Pólya) with limit $2/\pi$. The clique distances are checked exactly with a computer algebra system, and the bound is an exact fraction.

- `check2.py`, written separately, repeats the check with 60-digit floating point and its own sweep.
- `check3.py`, adapted from the checker of one of the referees, encloses g by Arb over whole intervals of t, bisecting until each enclosure is nonnegative, and builds the cliques and their radii in exact arithmetic of $\mathbb{Q}(\sqrt{3})$.

D	clique sets	certified bound for $m_D(\mathbb{R}^2)$	sweep	consequence
$\{1, 2\}$	$6 + 2$ triangles	$0.159222 < 1/6$	$[0, 1297]$	$\chi \ge 7$
$\{1, \sqrt{3}\}$	triangle, rhombus	$0.162866 < 1/6$	$[0, 757]$	$\chi \ge 7$
$\{1, (\sqrt{6} + \sqrt{2})/2\}$	none	$0.166018 < 1/6$	$[0, 107]$	$\chi \ge 7$
$\{1, 10\}$	none	$0.149599 < 1/6$	$[0, 36]$	$\chi \ge 7$
$\{1, 11/6\}$	none	$0.140995 < 1/7$	$[0, 171]$	$\chi \ge 8$
$\{1, 46/25\}$	none	$0.139185 < 1/7$	$[0, 175]$	$\chi \ge 8$
$\{1, 40\}$	none	$0.115511 < 1/8$	$[0, 36]$	$\chi \ge 9$
$\{1, 100\}$	none	$0.103302 < 1/9$	$[0, 37]$	$\chi \ge 10$
$\{1, 1000\}$	none	$0.089085 < 1/11$	$[0, 37]$	$\chi \ge 12$

TABLE 1. Certified density bounds (rounded up). For $\{1, 2\}$ the clique sets are six unit equilateral triangles and two equilateral triangles of side 2, each at one position; for $\{1, \sqrt{3}\}$, one unit equilateral triangle and one rhombus made of two such triangles (diagonals 1 and $\sqrt{3}$). All three programs accept every certificate.

D	certified bound for $m_D(\mathbb{R}^2)$	sweep	consequence
$\{1, 40, 1600\}$	$0.046389 < 1/21$	$[0, 36]$	$\chi \ge 22$
$\{1, 10, 100, 1000\}$	$0.039641 < 1/25$	$[0, 37]$	$\chi \ge 26$
$\{1, 100, 10^4\}$	$0.037168 < 1/26$	$[0, 37]$	$\chi \ge 27$
$\{1, 30, 900, 27000\}$	$0.021264 < 1/47$	$[0, 36]$	$\chi \ge 48$

TABLE 2. Certified density bounds for three and four distances (rounded up), without clique terms. `certify_lp2.py` and `check3.py` accept every certificate; the sweeps have up to $2.6 \cdot 10^7$ steps.

For $\{1, \sqrt{3}\}$ Oliveira Filho and Vallentin [22, §4] found the bound 0.170213 without clique terms, and 0.141577 for $\{1, j_{1,2}/j_{1,1}\}$, where $j_{1,k}$ are zeros of J_1; that ratio is presumably transcendental, so Theorem 12 does not apply to it directly, but nearby rationals such as 11/6 and 46/25 do. For the golden-ratio pair $\{1, (1 + \sqrt{5})/2\}$ of [24], the program with regular unit pentagons (whose sides and diagonals are 1 and $(1 + \sqrt{5})/2$) and triangles as cliques gives only 0.1769, above 1/6.

7. Uniform certificates for two distances

Let $W = \{(0, 0), (1, 0), (1/2, \sqrt{3}/2)\}$ be the unit equilateral triangle and, for a rational o, let $p_o = (1/2, \sqrt{3}/6 + o)$, the centroid of W moved by o along an axis of symmetry. The distances from p_o to the vertices of W are $r_1(o)$ (twice) and $r_3(o)$, where $r_1(o)^2 = 1/3 + o/\sqrt{3} + o^2$ and $r_3(o) = |1/\sqrt{3} - o|$. Let $M = \{0, 1, \omega, 1 + \omega, \rho, \rho\omega, \rho(1 + \omega)\} \subset \mathbb{C} = \mathbb{R}^2$, with $\omega = e^{i\pi/3}$ and $\rho = (5 + i\sqrt{11})/6$, be the Moser spindle: two rhombi, each made of two unit equilateral triangles, with the common vertex 0 and the far vertices $1 + \omega$ and $\rho(1 + \omega)$ at distance 1. Among any three of its points two are at distance 1, so a set avoiding the distance 1 contains at most two

of them. For $d \geq 1$, real numbers z_0, z_1, z_2, rationals $o_1, \dots, o_r$, points $q_1, \dots, q_n$ with rational coordinates and $y_c, u_m \geq 0$ put

$$S(t) = z_0 + z_1 J_0(t) + \sum_c y_c (2J_0(r_1(o_c)t) + J_0(r_3(o_c)t)) + \sum_m u_m \sum_{w \in M} J_0(|w - q_m|\, t),$$

$$g_d(t) = S(t) + z_2 J_0(dt),$$

and $s = z_0 + z_1 + z_2 + 3\sum_c y_c + 7\sum_m u_m$. Proposition 18, with the cliques W at the points p_{o_c} and the spindles M at the points q_m ($\alpha = 2$), shows: if $s > 0$ and $g_d \geq 0$ on $[0, \infty)$, then $m_{\{1,d\}}(\mathbb{R}^2) \leq (z_0 + \sum_c y_c + 2\sum_m u_m)/s$. Without triangles and spindles the bound is z_0/s. Besides $|J_0(x)| \leq \sqrt{2/(\pi x)}$ for $x > 0$ (Section 6) we use $|J_0'| = |J_1| \leq 1/\sqrt{2}$ and $|J_0''| = |J_0 - J_2|/2 \leq 1$ on $\mathbb{R}$ [11, 10.14.1], and $J_0 \geq -\mu$ on $[0, \infty)$ with $\mu = 0.4028$; the minimum of J_0 is $J_0(j_{1,1}) = -0.402759\ldots$, and the programs check $J_0 \geq -\mu$ in interval arithmetic on $[0, 4]$, while beyond 4 it follows from $\sqrt{2/(4\pi)} < 0.399$.

Lemma 19. *Let $D > 0$, $z_2 \geq 0$ and $s > 0$, and put $e(y) = \min\{\mu, \sqrt{2/(\pi y)}\}$ for $y > 0$. If*

$$h(t) = S(t) - z_2\, e(Dt) \;\geq\; 0 \qquad (t > 0),$$

then $g_d \geq 0$ on $[0, \infty)$ for every $d \geq D$, so that $m_{\{1,d\}}(\mathbb{R}^2) \leq (z_0 + \sum_c y_c + 2\sum_m u_m)/s$ for every $d \geq D$.

Proof. For $x \geq y > 0$ we have $J_0(x) \geq -\mu$ and $J_0(x) \geq -\sqrt{2/(\pi x)} \geq -\sqrt{2/(\pi y)}$, so $J_0(x) \geq -e(y)$. For $d \geq D$ and $t > 0$ apply this with $x = dt$ and $y = Dt$; since $z_2 \geq 0$, $g_d(t) \geq h(t) \geq 0$. Finally $g_d(0) = s > 0$. □

So one function serves every $d \geq D$, and checking it is a problem in one variable, in which nothing oscillates faster than $J_0(2.59\,t)$. Table 3 lists the eight certificates. Each is checked on $[0, T]$, with $T \in \{20, 40, 80\}$, and on $[T, \infty)$ by

$$h(t) \geq z_0 - |z_1|\sqrt{\tfrac{2}{\pi T}} - z_2\sqrt{\tfrac{2}{\pi DT}} - \sum_c y_c\Big(2\sqrt{\tfrac{2}{\pi r_1(o_c)T}} + \sqrt{\tfrac{2}{\pi r_3(o_c)T}}\Big) - \sum_m u_m \sum_{w \in M} \sqrt{\tfrac{2}{\pi |w - q_m| T}} > 0.$$

k	D	triangles	spindles	T	bound for all $d \geq D$	boxes
7	5.82	3	1	20	$0.1665379 < 1/6$	793
8	10.5	3	2	40	$0.1426634 < 1/7$	1209
9	18.9	3	2	80	$0.1249086 < 1/8$	1633
10	35.7	3	2	80	$0.1110473 < 1/9$	1843
11	73.6	2	2	80	$0.0999702 < 1/10$	2567
12	181	2	2	80	$0.0908619 < 1/11$	2171
13	655	1	3	80	$0.0833234 < 1/12$	4723
14	25 700	1	3	80	$0.0769214 < 1/13$	10263

TABLE 3. The certificates of Lemma 19 (bounds rounded up; `tail_kk.json`), each giving $\chi(\mathbb{R}^2, \{1, d\}) \geq k$ for every real algebraic $d \geq D$. The columns give the numbers of triangle positions o_c and of spindle points q_m used, with positive rational weights; the bound is $(z_0 + \sum_c y_c + 2\sum_m u_m)/s$, and the last column is the number of boxes of `tail.py` on $[0, T]$.

Remark. In floating point these thresholds are near the limit of the method (see the crossings after Theorem 5); for the program without configurations the limit is proved here. Consider first the program without triangles and spindles, $g_d(t) = z_0 + z_1 J_0(t) + z_2 J_0(dt)$. Put $\mu_0 = -\min J_0 = 0.40275\ldots$, let $d \geq 20$, and normalise $s = 1$. If $g_d \geq 0$, its values at $t = j_{1,1}/d$ and at a point t with $|t - j_{1,1}| \leq \pi/(2d)$ and $J_0(dt) = 0$ (there is one, because $dj_{1,1} > j_{0,1}$ and consecutive zeros of J_0 are less than π apart) give

$$z_0 + a z_1 - \mu_0 z_2 \geq 0 \qquad \text{and} \qquad z_0 + b z_1 \geq 0,$$

with $a = J_0(j_{1,1}/d) \in (0,1]$ and $|b+\mu_0| \le \eta = \pi^2/(8d^2) < \mu_0$, because $J_0'(j_{1,1}) = 0$ and $|J_0''| \le 1$. If $z_1 < 0$ the first inequality gives $z_0 > \mu_0/(1+\mu_0)$; otherwise the two give

$$z_0 \;\ge\; \frac{\mu_0(\mu_0-\eta)}{a+\mu_0+(1+\mu_0)(\mu_0-\eta)} \;>\; \frac{1}{12.2}.$$

As $d \to \infty$ the right-hand side tends to $L = \mu_0^2/(1+\mu_0)^2 = 1/12.13\ldots$, and L is the limit of the program: with μ_0 in place of μ in Lemma 19, the coefficients $z_1 = \mu_0/(1+\mu_0)^2$, $z_2 = 1/(1+\mu_0)$ and $z_0 = L + \delta$, where $\delta = z_2\sqrt{2/(\pi D)} + z_1(y_0/D)^2$ and $y_0 = 2/(\pi\mu_0^2)$, satisfy $h \ge 0$ and give the bound $(L+\delta)/(1+\delta)$ for every $d \ge D$. Indeed $L = \mu_0 z_1$ and $\mu_0 z_2 = (1+\mu_0)z_1$, so $h(t) = \delta + z_1(J_0(t)-1)$ for $Dt \le y_0$, $h(t) = \delta + z_1(\mu_0 + J_0(t) - (1+\mu_0)\sqrt{y_0/(Dt)})$ for $Dt \ge y_0$, and $J_0(t) \ge 1 - t^2/4$ for $t \le 2$ and $\mu_0 + J_0 \ge 0$ give $h \ge 0$ (separately for $Dt \le y_0$, for $y_0 \le Dt \le 2y_0$, for $2y_0 \le Dt$ with $t \le 1$, and for $t \ge 1$). Hence for $d \ge 20$ the program without triangles and spindles never gives more than 13 colours, and in floating point it never does for any d.

The same construction with $\mu_0 + J_0(t)$ replaced by $(1+\mu_0)Q(t)$, where Q is a certificate for one distance of the form of Proposition 18 with $Q(0) = 1$ and value v (that is, $z_0 + \sum_c \alpha_c y_c = v$), gives bounds that tend to $v\mu_0/(1+\mu_0)$: the case analysis needs only $Q \ge 0$, $Q(0) = 1$ and $Q(t) \ge 1 - Ct^2$, where C is the sum of $ar^2/4$ over the terms $aJ_0(rt)$ of Q with $a > 0$ (those with $a < 0$ only help, since $J_0 \le 1$), with the cut at $t = 1$ replaced by a cut at t_1 with $Ct_1^2 < 1 - 1/\sqrt{2}$. With our unit triangles v is about 0.271823, which explains the limit near 1/12.81 seen in floating point, so with triangles alone the program did not reach 14 colours. With spindles as well, v is about 0.262476 in floating point. A rigorous certificate with $v < 0.26789$ gives 14 colours for all large d in this way (with μ in place of μ_0): $S/S(0)$ for the certificate `tail_k14.json` is one, with $v = 0.267887$; Theorem 5 gives 14 colours from d_{14} on in any case. Fifteen colours would need $v < 0.2488$; the inclusion–exclusion program of [1] has $v = 0.24697$, which is how Proposition 20 below gives 15. For comparison, [19] reach (numerically) 0.26305 with spindles alone and 0.258795 with further inequalities.

For a bounded range of d we certify $g_d \ge 0$ for all d in an interval $[a,b]$ at once. On a box $[a,b] \times [t_0,t_1]$ the argument dt lies in $[at_0, bt_1]$, and J_0 on an interval $[m-r, m+r]$ is enclosed by $J_0(m) \pm (|J_1(m)|r + r^2/2)$, by $[-\mu, 1]$, and by $\pm\sqrt{2/(\pi x_0)}$ if the interval starts at $x_0 > 0$. The program bisects $[0,T]$ in t until every box has a nonnegative lower bound for g_d (with the triangle terms enclosed in the same way), and on $[T,\infty)$ it uses the envelope bound above, with $|z_2|\sqrt{2/(\pi aT)}$ for the term in d. Two files of such certificates, all with triangles, cover $[1.72, 2.04]$ with $k = 7$ (51 intervals) and $[1.82, 1.89]$ with $k = 8$ (14 intervals), which completes the proof of Theorem 5.

`tail.py` and `uniform.py` choose the coefficients by a floating-point linear program, round them to fractions, raise z_0 by a third of the slack, and prove $h \ge 0$, respectively $g_d \ge 0$ on each interval, with the enclosures above in Arb ball arithmetic; the same programs re-verify stored certificates, and `tail.py` reproduces its certificates byte for byte. `check_uniform.py`, written separately, recomputes every target $1/(k-1)$ from k, every radius $r_1(o)$, $r_3(o)$ from o and every spindle distance from the points of M (which it builds by complex multiplication), checks that the intervals of a file are contiguous and cover its range, encloses J_0 by point values and the Lipschitz bound $1/\sqrt{2}$ instead of the Taylor bound, bisects boxes in both t and d, sweeps h with steps chosen from its value, and checks $J_0 \ge -\mu$ itself. Both accept all ten certificates, and both reject corrupted ones: a lowered z_0, a scaled z_2, a negative or a missing triangle or spindle weight, a moved triangle or spindle, a threshold D moved below the certified one, a raised k, a gap between intervals, a reversed interval, an interval extended below its range, and a certificate whose g_d has minimum -10^{-5} at a point that a sampling scan misses.

Inclusion–exclusion at the unit scale. Ambrus, Csiszárik, Matolcsi, Varga and Zsámboki [1] bound $m_1(\mathbb{R}^2)$ by 0.2470 with a linear program on a set $X = \{x_1, \ldots, x_{23}\}$ of 23 points with coordinates in $\mathbb{Q}(\sqrt{3}, \sqrt{11})$: the probabilities of the patterns in which a random copy of X meets a set avoiding the distance 1 satisfy linear equations ("inclusion–exclusion"). We use their program at the scale of 1 and Lemma 19 at the scale of d. Let E_1 be the set of pairs $\{i,j\}$ with $|x_i - x_j| = 1$ (47 pairs), $\mathcal{S}$ the family of subsets of $\{1,\ldots,23\}$ containing no pair of E_1 (13 552

sets, including $\emptyset$), $r_{ij} = |x_i - x_j|$, and $\mathcal{C}$ a list of pairs (I, J) of sequences of distinct indices of the same length with $|x_{I_a} - x_{I_b}| = |x_{J_a} - x_{J_b}|$ for all a, b (so some isometry of the plane maps x_{I_a} to x_{J_a} for every a).

Proposition 20. *Let $D > 0$, $\delta_0 \in \mathbb{R}$ and $\mu \geq -\min J_0$, and put $e(y) = \min(\mu, \sqrt{2/(\pi y)})$. Suppose that real numbers $\psi_0, \psi_T, \lambda_i, u_{ij}$ ($\{i,j\} \notin E_1$), w_c ($c \in \mathcal{C}$) and $\eta, \tau \geq 0$ satisfy*

$$\Phi(t) = \psi_0 J_0(t) - \sum_i \lambda_i - \sum_{i<j} u_{ij} J_0(r_{ij} t) - \eta\, e(Dt) + 2\delta_0 \tau \;\geq\; 1 \qquad (t > 0),$$

$$\Phi_0 = \psi_0 - \sum_i \lambda_i - \sum_{i<j} u_{ij} + \eta + \tau(2\delta_0 - 1) \;\geq\; 1,$$

$$Q(S) = \psi_T + \sum_{i \in S} \lambda_i + \sum_{i<j,\ i,j \in S} u_{ij} + \sum_{(I,J) \in \mathcal{C}} w_{(I,J)}\big([I \subseteq S] - [J \subseteq S]\big) \;\geq\; 0 \qquad (S \in \mathcal{S}),$$

where the sums of u_{ij} run over the pairs $i < j$ with $\{i,j\} \notin E_1$. Then $m_{\{1,d\}}(\mathbb{R}^2) \leq \psi_T + \tau\delta_0^2$ for every $d \geq D$.

Proof. By Lemma 14 it suffices to bound the density δ of a periodic measurable set A, with period lattice Λ, avoiding 1 and d. Let ν be the spectral measure of its autocorrelation, as in the proof of Proposition 18, and κ the image of ν under $\xi \mapsto 2\pi|\xi|$: a measure on $[0, \infty)$ with total mass δ and $\kappa(\{0\}) = \delta^2$, and $f(r) = \int J_0(rt)\, d\kappa(t)$ is the radial average of the autocorrelation, so $f(1) = f(d) = 0$. Choose y uniformly in $\mathbb{R}^2/\Lambda$ and an isometry g of the plane fixing 0 uniformly (Haar measure on $O(2)$), and let $S = \{i : y + g x_i \in A\}$. Then $S \in \mathcal{S}$, $\Pr(i \in S) = \delta$ and $\Pr(i, j \in S) = f(r_{ij})$, and $\Pr(I \subseteq S) = \Pr(J \subseteq S)$ for $(I, J) \in \mathcal{C}$, because an isometry h with $h(x_{I_a}) = x_{J_a}$ changes the pair (y, g) into another uniformly distributed one. Taking expectations, $0 \leq \mathbb{E}\, Q(S) = \psi_T + \delta \sum_i \lambda_i + \sum_{i<j} u_{ij} f(r_{ij})$. Next, $f(d) = 0$, $J_0(0) = 1$ and $J_0(dt) \geq -e(dt) \geq -e(Dt)$ give $\kappa(\{0\}) \leq \int_{(0,\infty)} e(Dt)\, d\kappa$, and $\kappa(\{0\}) = \delta^2 \geq 2\delta_0\delta - \delta_0^2$. Integrating $\Phi \geq 1$ on $(0, \infty)$ and using $\Phi_0 \geq 1$ at 0,

$$\delta \leq \int_{(0,\infty)} \Phi\, d\kappa + \Phi_0 \kappa(\{0\}) = \psi_0 f(1) - \delta \sum_i \lambda_i - \sum_{i<j} u_{ij} f(r_{ij}) \\ - \eta\Big(\int_{(0,\infty)} e(Dt)\, d\kappa - \kappa(\{0\})\Big) + \tau\big(2\delta_0\delta - \kappa(\{0\})\big),$$

which is at most $\psi_T + \tau\delta_0^2$. □

Without the terms in η and τ, and with $\Phi \geq 1$ required at $t = 0$ as well, this is the bound $m_1(\mathbb{R}^2) \leq \psi_T$ of [1].

The hit set. At the scale of d Proposition 20 uses only $f(d) = 0$, and in floating point its value tends to about 0.0711 as $D \to \infty$ (end of this section), close to $0.2470\,\mu/(1+\mu) \approx 0.0709$. Heuristically: take a near-optimal solution of the one-distance program of [1], of value v, whose measure has mass at least v^2 at 0, as for genuine sets; mixing it with the empty pattern, with weights $\mu/(1+\mu)$ and $1/(1+\mu)$, and moving part of its mass at 0 to a point $t_a \to 0^+$ with $e(Dt_a) = \mu$ gives, up to $o(1)$, a solution of every constraint the proof uses, of value $v\mu/(1+\mu)$. We have not made this rigorous. A second program at the scale of d enters through the set of points near which A meets a copy of X. Put $n = |X|$ and $H(t) = \sum_{i,j=1}^n J_0(r_{ij} t)$ (with $r_{ii} = 0$). Then $H(t) = \int_{S^1} \big|\sum_{j=1}^n e^{it\langle \theta, x_j \rangle}\big|^2 d\sigma(\theta)$, where σ is the uniform probability measure on the unit circle, so $0 \leq H \leq n^2$.

Lemma 21. *Let $\beta_0, \beta_T, b_i, v_{ij}$ ($\{i,j\} \notin E_1$), ϖ_{ij} ($\{i,j\} \in E_1$) and ω_c ($c \in \mathcal{C}$) be real numbers with*

$$\Phi_B(t) = \beta_0 J_0(t) - \sum_i b_i - \sum_{i<j} v_{ij} J_0(r_{ij} t) \;\geq\; 1 \qquad (t \geq 0)$$

and $Q_B(R) = \beta_T + \sum_{i \in R} b_i + \sum_{i<j,\ i,j \in R} v_{ij} + \sum_{\{i,j\} \in E_1(R)} \varpi_{ij} + \sum_{(I,J) \in \mathcal{C}} \omega_{(I,J)}\big([I \subseteq R] - [J \subseteq R]\big) \geq 0$ for every *$R \subseteq \{1, \ldots, n\}$, where the sums of v_{ij} run over the pairs $i < j$ with $\{i,j\} \notin E_1$ and $E_1(R)$ is the set of pairs of E_1 inside R, and put $L_B = \max(0, \beta_0 + \sum_{\{i,j\} \in E_1} \varpi_{ij})$. Let*

$(B_g)_{g\in O(2)}$ be Λ-periodic sets such that $(g,y)\mapsto 1_{B_g}(y)$ is measurable, q the average of their densities, and ε the average of the radial averages of their autocorrelations at distance 1, both averages over the Haar measure of $O(2)$. Then $q\le\beta_T+L_B\varepsilon$.

Proof. Choose g, $y\in\mathbb{R}^2/\Lambda$ and $h\in O(2)$ uniformly and let $R=\{i : y+hx_i\in B_g\}$. As in the proof of Proposition 20, $\Pr(i\in R)=q$, $\Pr(i,j\in R)=f_B(r_{ij})$ with $f_B(r)=\int J_0(rt)\,d\kappa_B(t)$ for the average κ_B of the spectral measures (total mass q), $\Pr(i,j\in R)=\varepsilon=f_B(1)$ for $\{i,j\}\in E_1$, and the congruence relations hold. Now R may contain pairs of E_1, but $Q_B(R)\ge 0$ for every R, so $0\le\mathbb{E}\,Q_B(R)=\beta_T+q\sum b_i+\sum v_{ij}f_B(r_{ij})+\varepsilon\sum\varpi_{ij}$, while $q\le\int\Phi_B\,d\kappa_B=\beta_0\varepsilon-q\sum b_i-\sum v_{ij}f_B(r_{ij})$. Adding, $q\le\beta_T+(\beta_0+\sum\varpi_{ij})\varepsilon\le\beta_T+L_B\varepsilon$, since $\varepsilon\ge 0$: the autocorrelation of an indicator function is nonnegative. □

Proposition 22. *Let the hypotheses of Proposition 20 hold, except that Φ and $Q(\emptyset)$ are replaced: for some $\zeta\ge 0$ and numbers as in Lemma 21,*

$$\Phi(t)-\zeta L_B\,e(Dt)(n^2-H(t))\ \ge\ 1\quad(t>0),\qquad Q(S)\ge 0\quad(\emptyset\ne S\in\mathcal{S}),\qquad \psi_T\ge\zeta.$$

Then $m_{\{1,d\}}(\mathbb{R}^2)\le\psi_T+\tau\delta_0^2-\zeta(1-\beta_T)$ for every $d\ge D$.

Proof. Keep the notation of the proof of Proposition 20, and let $q=\Pr(S\ne\emptyset)$, the average over g of the density of $B_g=\{y : (y+gX)\cap A\ne\emptyset\}=\bigcup_i(A-gx_i)$. Scaled by $1/d$, these sets satisfy Lemma 21 with $\varepsilon=\mathbb{E}_g\,f_{B_g}(d)$. Since $1_{B_g}(y)\le\sum_i 1_A(y+gx_i)$,

$$\begin{aligned}\varepsilon\le\sum_{i,j}\mathbb{E}_{g,v}\,\varphi(dv+g(x_j-x_i))&=\int J_0(dt)H(t)\,d\kappa(t)\\&=\int_{(0,\infty)}J_0(dt)\big(H(t)-n^2\big)\,d\kappa(t)\le\int_{(0,\infty)}e(Dt)\big(n^2-H(t)\big)\,d\kappa(t),\end{aligned}$$

where v is uniform on the unit circle, φ is the autocorrelation of A, the averages over v and g of $e^{2\pi i\langle\xi,dv+gw\rangle}$ are $J_0(2\pi d|\xi|)J_0(2\pi|\xi||w|)$, the middle equality subtracts $n^2f(d)=0$, and the last inequality uses $J_0(dt)\ge -e(dt)\ge -e(Dt)$ and $H\le n^2$. So $1-\Pr(S=\emptyset)=q\le\beta_T+L_B\varepsilon$, and $\mathbb{E}\,Q(S)\ge\zeta\Pr(S=\emptyset)$ gives $\psi_T+\delta\sum\lambda_i+\sum u_{ij}f(r_{ij})\ge\zeta(1-\beta_T-L_B\varepsilon)$. Integrating as before, now with the extra term, $\delta\le\psi_T-\zeta(1-\beta_T-L_B\varepsilon)+\tau\delta_0^2-\zeta L_B\int_{(0,\infty)}e(Dt)(n^2-H)\,d\kappa\le\psi_T+\tau\delta_0^2-\zeta(1-\beta_T)$. □

As $D\to\infty$ the term with ζ tends to 0 uniformly in t, since $\sup_{t>0}e(Dt)\big(n^2-H(t)\big)=O(D^{-1/2})$ (use $n^2-H(t)\le t^2\sum_{i,j}r_{ij}^2/4$ for small t). So the certificate of Lemma 21, multiplied by $1+O(D^{-1/2})$, can serve as well on the A side, with $\eta=\tau=0$ and $\zeta=\psi_T$; this gives the bound $(1+O(D^{-1/2}))\,\beta_T^2$, so $\limsup_{d\to\infty}m_{\{1,d\}}(\mathbb{R}^2)\le\beta_T^2$. In floating point the value of the program of Proposition 22 approaches about 0.0610, close to β_T^2. At the scale of d the set A behaves like a set avoiding d whose points are, at the scale of 1, spread like a set avoiding 1. Proposition 22 is thus an effective form of the upper bound in Bukh's theorem $m_{\{1,d\}}(\mathbb{R}^2)\to m_1(\mathbb{R}^2)^2$ [2], with $m_1(\mathbb{R}^2)$ replaced by the inclusion–exclusion bound β_T and with usable thresholds; Bukh remarks that his zooming-out and supersaturation lemmas are effective, but states no threshold.

The certificates for 15, 16 and 17 colours. The linear programs behind Propositions 20 and 22 have the inclusion–exclusion rows of [1] for X (13 552 atoms; we generate the congruences ourselves: the independent sets with at least two points fall into 11 226 congruence classes, which give 2 302 relations $\Pr(I\subseteq S)=\Pr(J\subseteq S)$ and imply all others), the variables κ on a grid of t, and the rows for $f(1)=0$, the envelope, the tangent and the hit set. We solve them in floating point, round the multipliers to fractions, and verify the hypotheses exactly or in interval arithmetic. Table 4 lists the certificates.

Two programs check them. `check_ie.py` computes in $\mathbb{Q}(\sqrt{3},\sqrt{11})$ the unit pairs of X, the independent sets and every congruence, checks $Q\ge 0$ on the atoms in exact arithmetic, checks $Q_B\ge 0$ on all 2^{23} subsets of X with integers over a common denominator, and proves $\Phi\ge 1$ (with the hit-set term) and $\Phi_B\ge 1$ by bisecting $[0,t_{\max}]$ in t with the Taylor enclosures of J_0 above, in Arb ball arithmetic, and by $|J_0(x)|\le\sqrt{2/(\pi x)}$ beyond $t_{\max}$. `check_ie_indep.py`,

k	D	certificate	bound for all $d \geq D$	proved by
15	$5 \cdot 10^5$	`ie_k15.json`	$0.0713614 < 1/14$	Proposition 20
16	$7 \cdot 10^7$	`ie_k16.json`	$0.0657685 < 1/15$	Proposition 22
17	$8 \cdot 10^8$	`ie_k17.json`	$0.0623330 < 1/16$	Proposition 22

TABLE 4. The inclusion–exclusion certificates (bounds rounded up), each giving $\chi(\mathbb{R}^2, \{1, d\}) \geq k$ for every real algebraic $d \geq D$. The last two use the certificate `ie_B.json` of Lemma 21, with $\beta_T = 0.2470097$ and $L_B = 17.764$.

written separately, uses its own interval arithmetic on integers, its own power series and Hankel expansion for J_0, rigorous values of π and of the distances, and a sweep with steps chosen from the values and the bound $|J_0''| \leq 1$; it has its own transcription of the coordinates of X, checked against the notebook of [1]. Both accept the four certificates and reject corrupted ones (a broken congruence, a lowered or misplaced multiplier, a raised η, ζ or τ, a lowered penalty of the B side). In floating point the program of Proposition 20 tends to about $0.0711 = 1/14.06$ as $D \to \infty$, close to $0.24697 \cdot \mu/(1+\mu)$, so it gives no more than 15 colours; the program of Proposition 22 gives $0.065758 = 1/15.21$ at $D = 7 \cdot 10^7$ and $0.062324 = 1/16.05$ at $D = 8 \cdot 10^8$, and tends to about $0.0610 = 1/16.4$, close to 0.24697^2, so it gives no more than 17.

8. Remarks

(1) The hard input is OpenAI's rigidity theorem [23, Thm. 2.3], which OpenAI formalized in Lean. Our contribution is the observation that their transfer holds for every set of algebraic distances and for fractional colourings, the consequences of combining it with density bounds, and the two-scale argument of Section 5, which uses OpenAI's theorem a second time, as a statement about finite graphs.

(2) Davies [7] proves, by a different method, that in $\mathbb{R}^d$ with $d \geq 4$, and for finite D whose elements have rational squares, the chromatic number equals the measurable chromatic number and the independence ratio equals the measurable one; his certificates come from the completely positive framework of DeCorte, Oliveira Filho and Vallentin and are transferred through lattice embeddings. With Bukh's theorem this gives exponential growth in k for $d \geq 4$, and he hopes to extend the method to $d = 2, 3$. Our results concern $d = 2$ (and every $d \geq 2$ for Corollaries 6 and 10, by restriction to a plane), allow arbitrary algebraic distances, and rest on OpenAI's transfer.

(3) The distances must be algebraic because [23, Thm. 2.3] is a statement about the algebraic plane. This is a limitation of the method, but probably also of the statement: if Bukh's conjecture above holds, the inequality $\chi(\mathbb{R}^2, D) \geq 1/m_D(\mathbb{R}^2)$ fails for some algebraically independent D.

(4) No explicit graph is given. The graph of Dúcz and Varga can in principle be made explicit but is astronomically large; an explicit unit-distance graph of reasonable size with fractional chromatic number above 4, or an explicit graph with edges of lengths 1 and 2 and no proper 6-colouring, remains to be found.

References

[1] G. Ambrus, A. Csiszárik, M. Matolcsi, D. Varga and P. Zsámboki, *The density of planar sets avoiding unit distances*, Math. Program. 207 (2024) 303–327; arXiv:2207.14179.

[2] B. Bukh, *Measurable sets with excluded distances*, Geom. Funct. Anal. 18 (2008) 668–697.

[3] T. Bellitto, A. Pêcher and A. Sédillot, *On the density of sets of the Euclidean plane avoiding distance* 1, Discrete Math. Theor. Comput. Sci. 23(1) (2021), doi:10.46298/dmtcs.5153; arXiv:1810.00960.

[4] A. Chybowska-Sokół, K. Junosza-Szaniawski and K. Węsek, *Coloring distance graphs on the plane*, arXiv:2201.04499.

[5] H. T. Croft, K. J. Falconer and R. K. Guy, *Unsolved Problems in Geometry*, Springer, 1991.

[6] H. T. Croft, *Incidence incidents*, Eureka 30 (1967) 22–26.

[7] J. Davies, *Measurable obstructions for unmeasurable colourings*, arXiv:2610.12301 (8 October 2026).

[8] N. G. de Bruijn and P. Erdős, *A colour problem for infinite graphs and a problem in the theory of relations*, Indag. Math. 13 (1951) 369–373.

[9] A. D. N. J. de Grey, *The chromatic number of the plane is at least* 5, Geombinatorics 28 (2018) 18–31; arXiv:1804.02385.
[10] J. De Neve, F. Vanden Kerchove, D. Colle, W. Tavernier and M. Pickavet, *On the chromatic number of the plane with two forbidden distances*, Amer. Math. Monthly 132 (2025).
[11] NIST Digital Library of Mathematical Functions, `https://dlmf.nist.gov`, equation 10.14.1.
[12] Á. Dúcz and D. Varga, *A unit-distance graph in the plane with independence ratio below* 1/4, arXiv:2606.28157 (26 June 2026).
[13] G. Exoo and D. Ismailescu, *A* 6*-chromatic two-distance graph in the plane*, arXiv:1909.13177.
[14] T. F. Bloom, *Erdős Problem #706*, `https://www.erdosproblems.com/706`, accessed 10 October 2026.
[15] P. Erdős, *On the combinatorial problems which I would most like to see solved*, Combinatorica 1 (1981) 25–42.
[16] P. Erdős, *Some combinatorial and metric problems in geometry*, in: Intuitive Geometry (Siófok, 1985), Colloq. Math. Soc. János Bolyai 48, North-Holland, 1987, 167–177.
[17] D. Geller and S. Stahl, *The chromatic number and other functions of the lexicographic product*, J. Combin. Theory Ser. B 19 (1975) 87–95.
[18] J. Grytczuk, K. Junosza-Szaniawski, J. Sokół and K. Węsek, *Fractional and j-fold colouring of the plane*, Discrete Comput. Geom. 55 (2016) 594–609; arXiv:1506.01887.
[19] T. Keleti, M. Matolcsi, F. M. de Oliveira Filho and I. Z. Ruzsa, *Better bounds for planar sets avoiding unit distances*, Discrete Comput. Geom. 55 (2016) 642–661; arXiv:1501.00168.
[20] M. Matolcsi, I. Z. Ruzsa, D. Varga and P. Zsámboki, *The fractional chromatic number of the plane is at least 4*, arXiv:2311.10069 (v4, 2025).
[21] E. Naslund, *The chromatic number of* $\mathbb{R}^n$ *with multiple forbidden distances*, Mathematika 69 (2023); arXiv:2205.12312.
[22] F. M. de Oliveira Filho and F. Vallentin, *Fourier analysis, linear programming, and densities of distance avoiding sets in* $\mathbb{R}^n$, J. Eur. Math. Soc. 12 (2010) 1417–1428.
[23] OpenAI, *The Euclidean plane is not five-colorable*, preprint (23 September 2026), with a formalization in Lean, `https://github.com/openai/math`.
[24] J. Owings, M. Tetiva and M. Huddleston, *Coloring the plane*, Amer. Math. Monthly 115(2) (2008) 170–172.
[25] J. Parts, *A small* 6*-chromatic two-distance graph in the plane*, Geombinatorics 29(3) (2020) 111–115; arXiv:2010.12656.
[26] D. H. J. Polymath, *The Hadwiger–Nelson problem*, Polymath16 project, 2018–2020.
[27] S. Galán, *The Hadwiger–Nelson problem over number fields*, code, data and formal proofs, 2026. Archived at Zenodo, doi:10.5281/zenodo.22976635 (all versions); source at `https://github.com/decalion89/chromatic-number-of-the-plane`.
[28] A. Soifer, *The New Mathematical Coloring Book*, Springer, 2024.